Branch-specific rate estimation (R2.2) and error vs trend (R2.3)

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Supplementary analyses addressing Reviewer 2, comments R2.2 and R2.3.

- **R2.2**: Branch-specific rate estimation errors from the OBDD calibration study.
- **R2.3**: Estimation error as a function of the trend parameter.

Uses only the existing MCMC parameter traces from the calibration study (no re-simulation or posterior tree files needed).

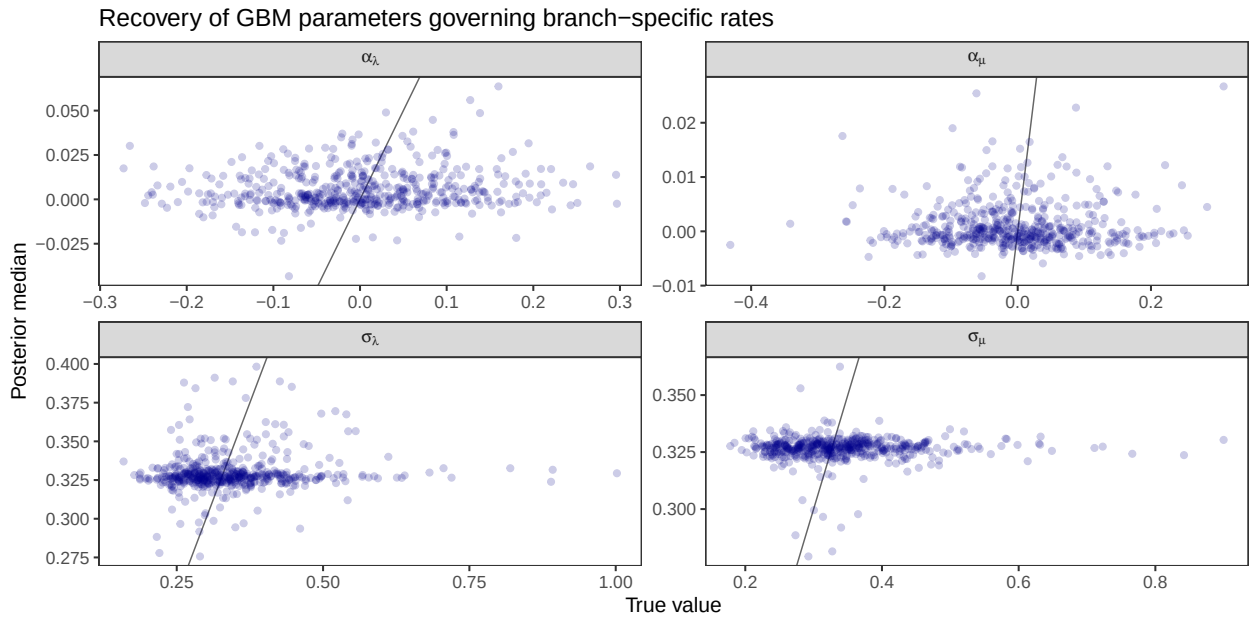
Load calibration data

```
## Number of datasets: 500
```

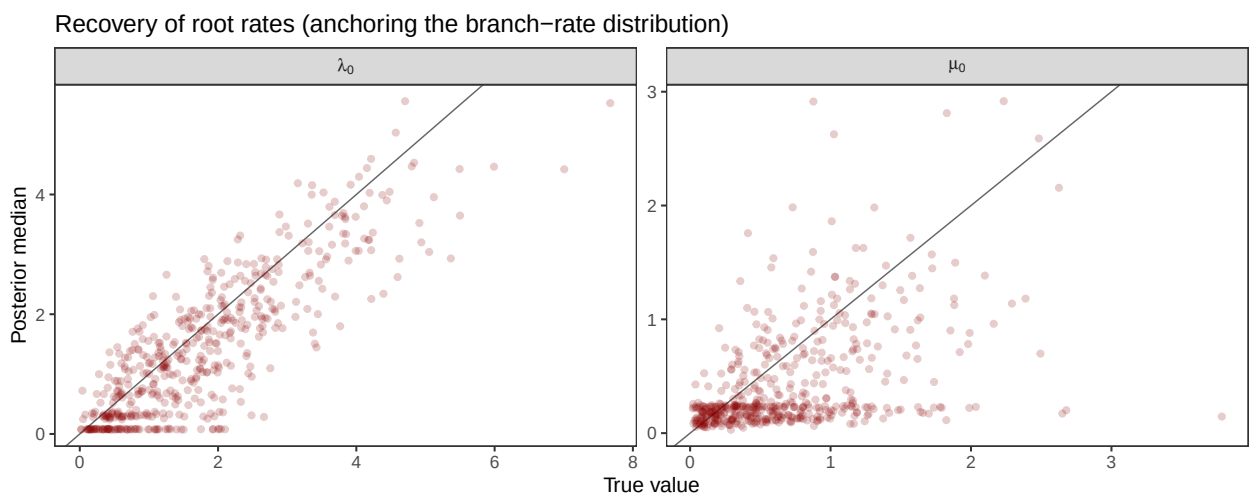
R2.2: Branch-specific rate estimation

In the OBDD model, branch-specific rates evolve according to a geometric Brownian motion parameterised by the root rates (λ_0, μ_0), trend parameters (α_λ, α_μ), and diffusion parameters (σ_λ, σ_μ). These six parameters **fully determine** the distribution of branch-specific rates across the tree. If they are accurately recovered, the implied branch-rate distribution is accurately recovered.

Recovery of GBM parameters

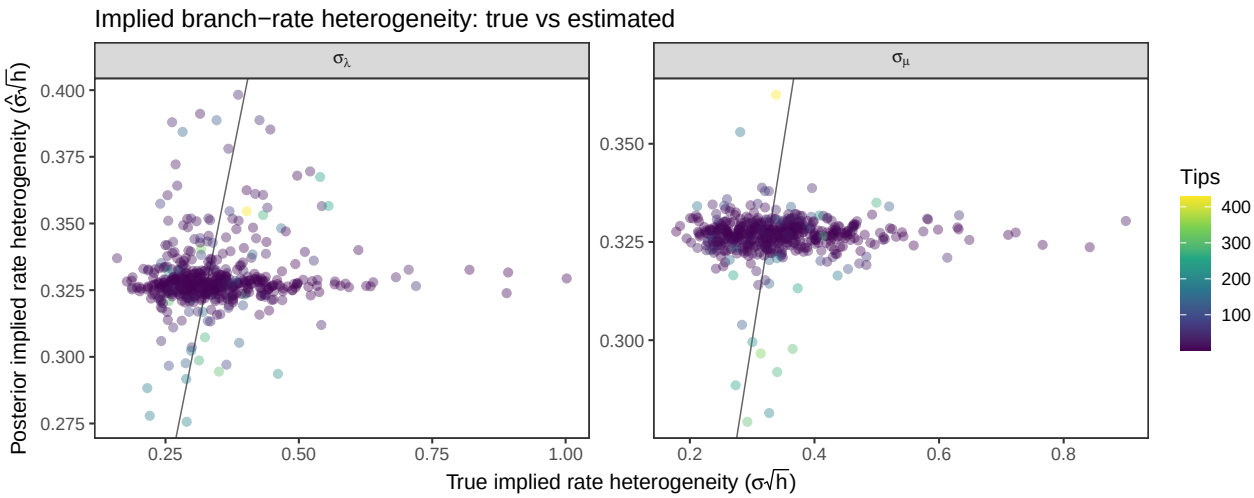


Root rate recovery

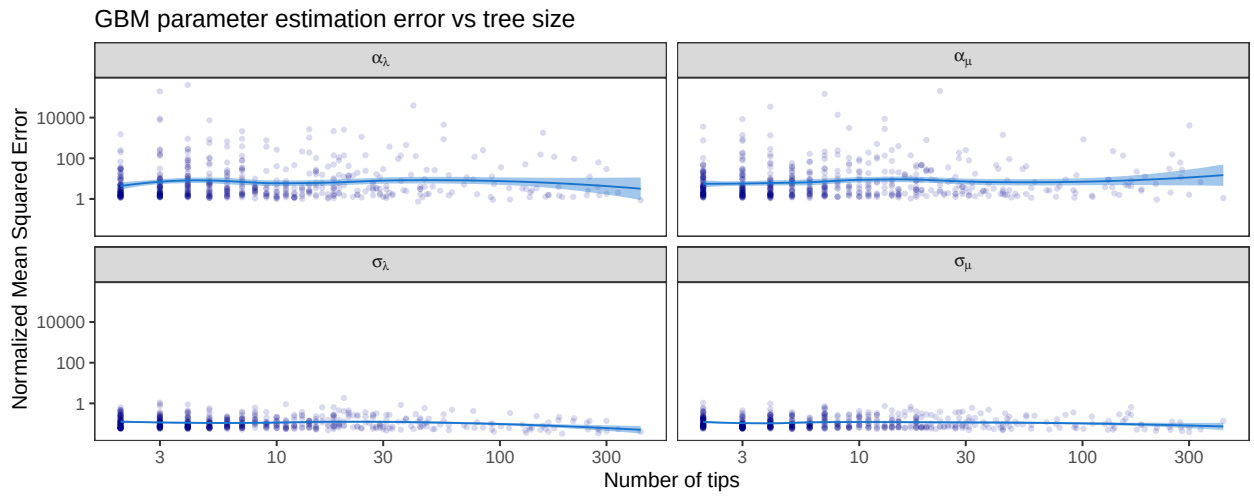


Implied rate heterogeneity

Under the GBM, the variance of log-rates across branches at time t from the root is $\sigma^2 t$. For a tree of height h, the expected standard deviation of log-λ across lineages at the tips is approximately $\sigma_\lambda \sqrt{h}$. We can compare the implied heterogeneity from the true vs. posterior σ estimates.



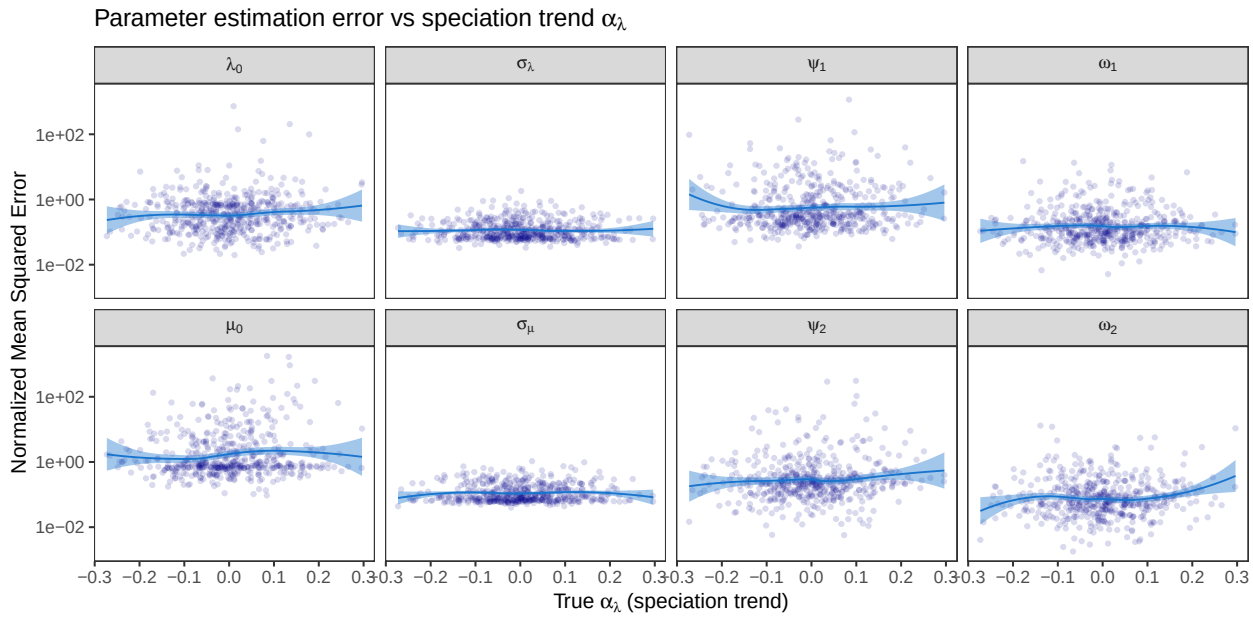
NMSE of GBM parameters vs tree size



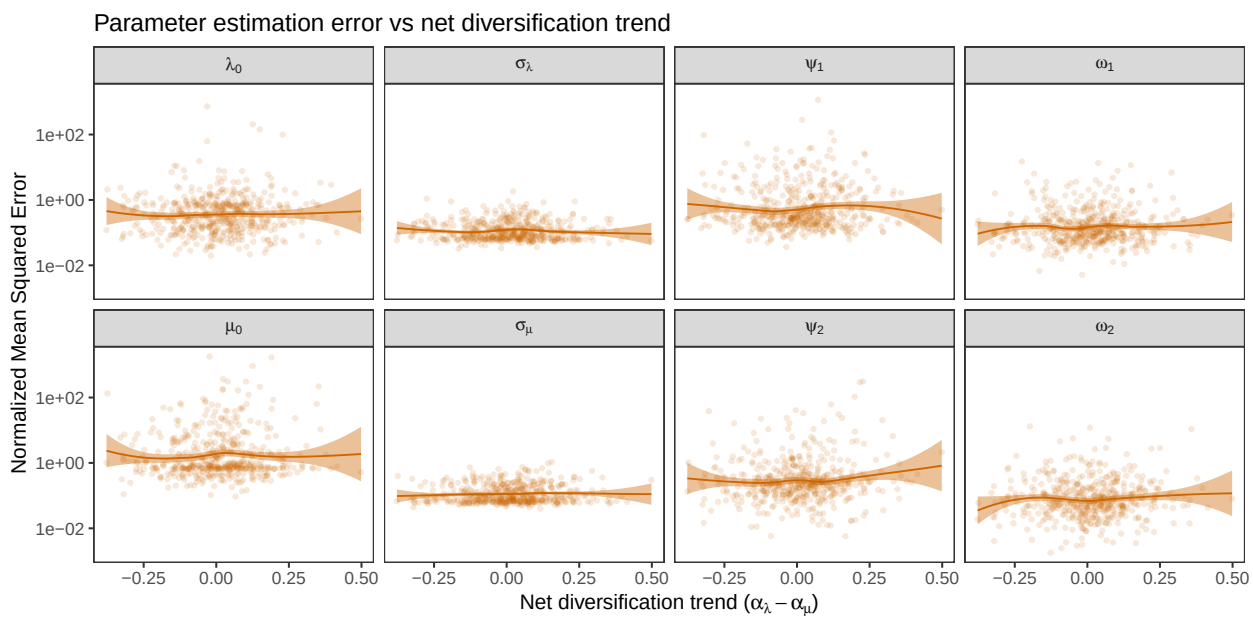
R2.3: Estimation error vs trend parameter

The reviewer predicts that in models with a positive net-diversification trend ($\alpha_\lambda > 0$ or $\alpha_\mu < 0$), the estimation error should be smaller because the tree is expected to be larger and more informative.

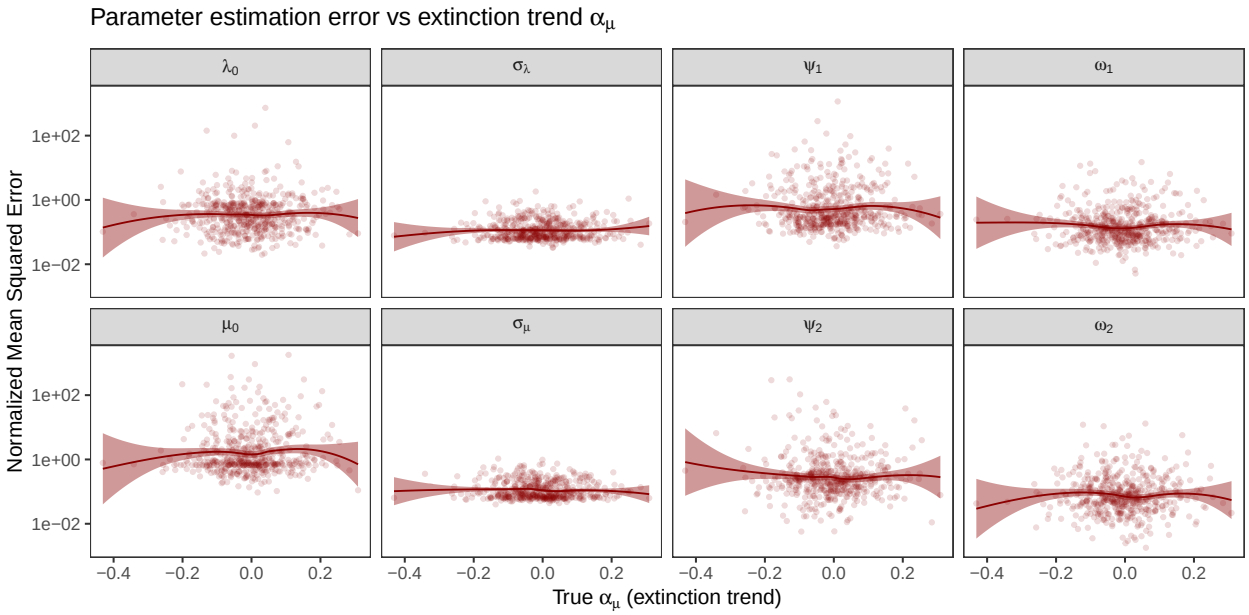
NMSE vs speciation trend α_λ

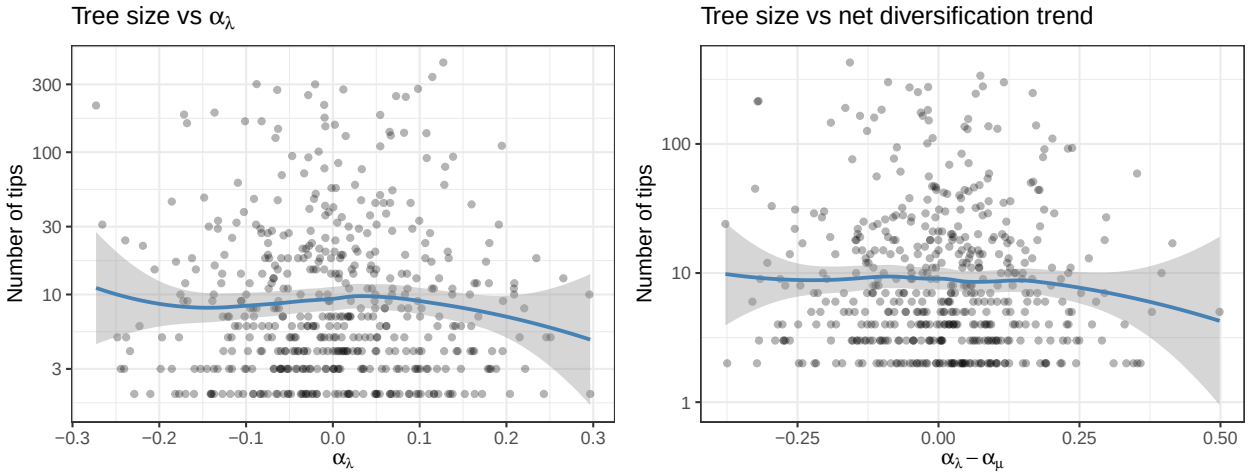


NMSE vs net diversification trend $\alpha_\lambda - \alpha_\mu$



NMSE vs extinction trend α_μ



Tree size as confound

The reviewer's hypothesis is that positive net-diversification trends produce larger trees, which provide more information. We verify this expected confound.